Nudging Justice: Uncovering Behavioural Chains at the Gandhinagar District Court

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I. ABSTRACT

Abstract - Witnesses play a vital role as determining factors in the outcome of proceedings in the justice system prevalent in India. With 5.58 crore pending cases in the Indian courts, lakhs of witnesses are involved in the ongoing cases across the country. Due to the procedural complexity and a lack of general guidelines, institutional anxiety in witnesses is a common phenomenon. To combat this, thorough ethnography and user interviews were conducted uncovering behaviour chains validated through simulations. Solutions which are low-cost, and low-effort that could lead to a significant positive improvement in witness demeanor have been proposed and discussed in this paper.

II. THE FRAME

The Indian Judicial System was established in 1950 with the Supreme Court as the highest authority to uphold the rule of law and define societal deviances. Indian Courts are known as “Open Courts” under the belief that justice should not only be served but must also be witnessed as serving by the society. The system is a unified and independent body with a set 3-level hierarchy. Supreme Court is the apex, followed by state High Courts, district Subordinate Courts and special courts such as the Family and Consumer Courts.

As of March 2026, and based on recent data from National Justice Data Grid (NJDG), a significant 30% of all civil disputes develop as contested issues requiring the presence of witnesses during the contested proceedings. Contested family court cases typically last for 2 to 5 years due to uncooperative witnesses, frequent appeals and delayed evidence. Based on recent data sourced from the National Judicial Data Grid (NJDG), 11.95 Lakh cases are pending in Family Courts across all Indian states. The stipulated timelines for resolving matrimonial cases are within one year and for maintenance cases, 90 days. Reasons for pending cases in Family Courts involve frequent adjournments, insufficient evidence, procedural delays, witness compromise and unfair proceedings, medical reasons and absentees.

This paper explores the kinds of witness behaviours exhibited in the court when they are more often than not unaware of the courts’ complex functioning, Additionally, this paper also lists the various triggers for situational behaviours, how such similar behaviours are manifested in multiple spatial contexts, the importance of addressing these behaviours in most contexts and subtle yet low-cost solutions to address the same.

III. THE LANDSCAPE

Study Area - The District Court of the District of Gandhinagar, Gujarat was the chosen context in order to understand the behaviours of three primary stakeholders:

1. Advocates and judges
2. Court staff officials
3. Public/Litigants/Witnesses

The District Court of Gandhinagar addresses both civil and criminal cases. Family Courts and Consumer Courts fall under the respective State’s High Court jurisdiction as a Special Court and are

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located in the same premises as the District Court. Family Courts have been established to promote conciliation in and secure speedy settlements of disputes regarding marriage and other family affairs. These courts are the specialized starting points for all matters related to marriage and families.

Our focus pivoted towards deeply observing and uncovering behaviours in the following spaces of Family Court:

1. Waiting Area

2. Family Courtroom Witness Box

Research Aim - The design research was conducted to understand the influence of spaces and how environment trigger various human behaviours initially in different stakeholders and later narrowing down to witnesses in the context of Family Courts. This is particularly significant given the magnitude of negative impact that any witness compromise has on the fairness of proceedings. Understanding the types of emphasized behaviours would further aid in defining a societal line of reference or ideal behaviour at the Family Court and help in classifying a particular behaviour as deviant from normal. An important revelation emerging from the research is that deviant behaviours that are socially accepted cannot be normalized. This research was also intended to bring about sustainable long-term behaviour changes to enhance the efficiency and smooth the functioning of the Family Court through minute and refined solutions.

Research Methodologies - The design research commenced with slow ethnography and learning to incorporate visual thinking while observing a plethora of human behaviours in the three chosen spaces of the Family Court. Unstructured and structured user interviews were conducted to aid us in validating our assumptions and triggers for behaviour manifestation and to understand the demographics of the litigants at the Family Court. Emphasis was placed on noting overt behaviours, differentiating between overt and covert behaviours and measuring behaviours in terms of the number of people displaying them, the frequency of behaviour display and the impact of display of behaviour. Both positively and negatively deviant behaviours were observed. Negative deviant behaviours were given a larger prominence as the research aim was also to increase the efficacy of the court. Behavioural chains such as the situational witness behaviours and crowding around the courtroom doorway were discovered during the process.

Simulations in the form of rapid fire/memory based quizzes with design students and university staff were designed and attempted to replicate the emerging witness behaviours and validate research assumptions. Findings from varying the dependent variables in the first set of simulations were then applied to ideate solutions to curb the situational behaviours displayed by the witness. Second round of simulations with low fidelity prototypes was designed to understand the efficacy of ideated solutions.

IV. THE TURN

The Doorway Bottleneck - The structured interviews and simulations provided deepened initial views of how litigants see the judicial system. Courtrooms are perceived as institutions of fear rather than as institutions of authority, protection and justice. With this prevailing perception, the lack of information in vernacular languages, the prevalence of legal jargon and reliance on advocates as the sole source of any information regarding their case can bring about a sense of apathy in the litigants. These interruptions and distractions by the crowd could also have direct effects of cognitive fatigue, overload and loss of focus on both the witnesses in the cross examination and the witnesses seated in the waiting area of the Family Court.

Over a span of 3 hours, 78 persons were observed to be standing at the doorway of the courtroom, with their eyes directed towards the dias of the courtroom. 83 people entered, exited and obstructed movement in and out. Right next to the courtroom doorway, 22 people standing at arm's length from a single cause list booklet were seen looking at it with an average of a 4-people cluster at every 15 minute interval, over a period

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of 120 minutes. 10 people were seen tilting their heads at the list over a span of 90 minutes with 11 people spending more than 1 minute in front of the list. 17 people were observed holding the booklet with both hands. Refer Fig.1. Mapping of Doorway Bottleneck behaviour across time, space and behaviour and Fig.2. Mapping Doorway Bottleneck cause list behaviour

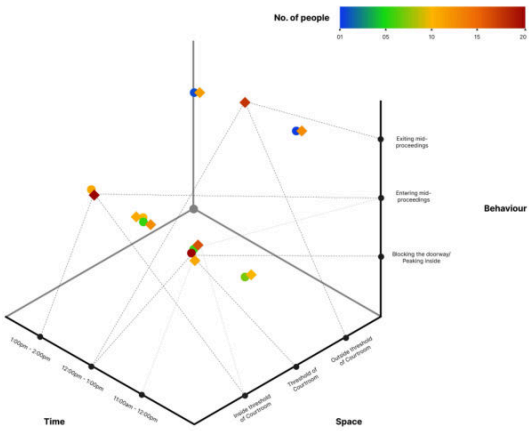


Fig.1. Mapping Doorway Bottleneck behaviour across time, space and behaviour

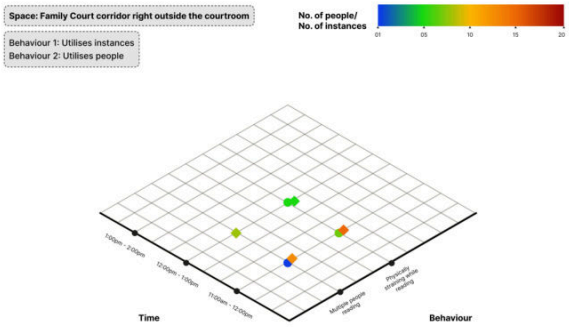


Fig.2. Mapping Doorway Bottleneck cause list behaviour

The Family Court of the Ahmedabad District and the High Court of the State of Gujarat display real-time case status and information regarding the order of the cases which are to be heard next for the given day. This automated information system has been replaced by a notice board and bundled information notebooks next to the doorway at The Family Court of the Gandhinagar District. This information system prompted the advocates to stand in the vicinity of the doorway awaiting their turns. Litigants, due to lack of available information regarding their conduct, were observed tailing their advocates closely in the waiting area of the Family Court. Given the

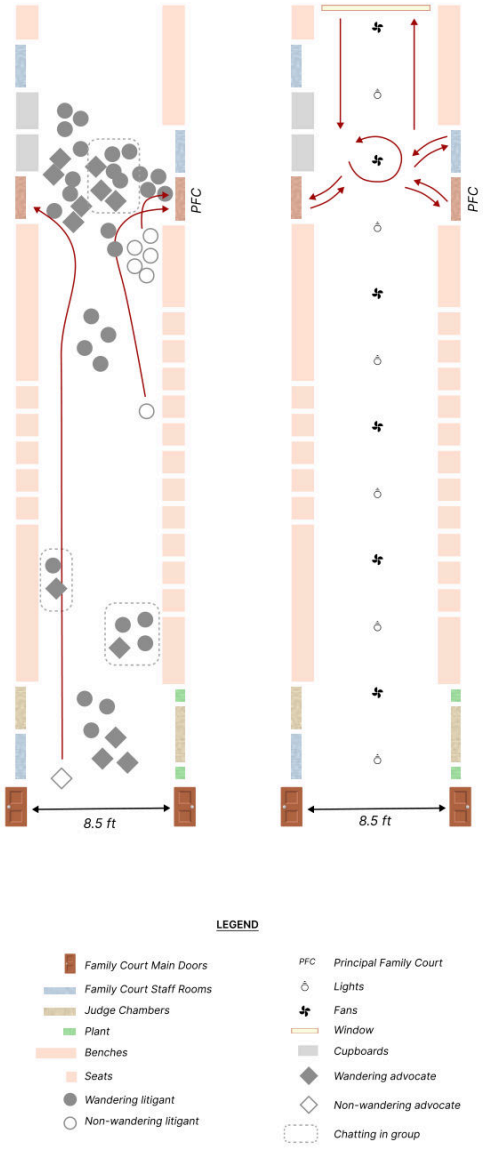


Fig.3a. Mapping of Litigants and Advocates in the Family Courtroom Corridor Fig.3b. Family Courtroom ventilation paths

architecture of the Family Court floor, the doorway of the Courtroom is also the point of maximum airflow providing optimal cross ventilation prompting crowding. *Refer Fig.3a. Mapping of Litigants and Advocates in the Family Courtroom Corridor and Fig.3b. Family Courtroom ventilation paths*

The bundled cause list booklet contains a day wise list of targeted cases, family suits and criminal cases along with the case number, the opposing parties, the names of advocates, the state of the case printed in a table format only in English language. Gujarati was concluded to be

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the primary language of the litigants through structured interviews. Lack of accessible information coupled with the method of display of the cause list caused further doorway crowding and passive participation from litigants in their own cases.

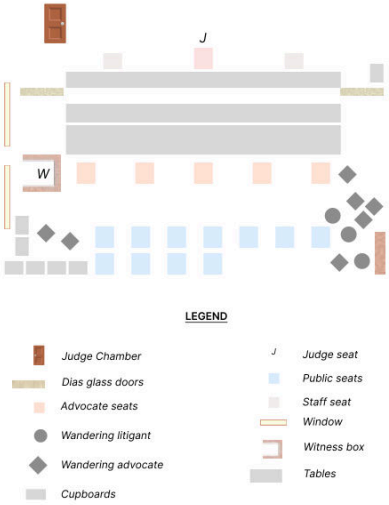


Fig.4. Map of Principal Family Court

The Situational Witness Behaviour *- While standing in the witness box and during cross questioning, 3 witnesses moved their heads toward the public seating area in the courtroom and directed their eyes toward the individuals seated there. One witness produced a verbal statement in response to a question, and then within 5 seconds responded with a second verbal statement to the same question that differed from the first. While the witness was being questioned, someone seated in the public area spoke in an audible volume after which the witness directed their eyes toward them. This scenario was observed twice in different cases. Refer Fig.4. Map of Principal Family Court*

The witness box in India is positioned such that the witness is able to view the presiding Judge, cross examining advocate and the public seated in the court. Given the high stakes in courts, and very frequent interruptions disrupting the decorum of the court, witnesses often fall back to using their bounded rationality to answer. Due to the stretched out timelines, recalling events from a distant memory can lead to memory distortion and inconsistent statements. Family suits being personal issues could also harbor a fear of being judged socially in the witness due to the open nature of courts. For a number of first-time witnesses, entering a witness box is a completely new and uneasy experience due to prior conceived notions about the space. This renders it easy to compromise a witness and tutor them answers through visual/verbal cues proving to be catastrophic. Remaining standing during the complete length of the cross-examination, as a mark of solemnity and seriousness, has also been observed to be uncomfortable, costing the witness their focus and attention.

The above interpretations referring to witness behaviours have been validated through behaviour replication simulations leading one to question why witnesses deemed the “eyes and ears of justice”, essential for proving cases, are not optimally cared for and protected. Any person is allowed to testify as a witness given their competency and witness examination is governed by the Bharatiya Sakshya Adhiniyam 2023 which replaced the Indian Evidence Act of 1872. But due to insufficient infrastructure and concrete guidelines, witnesses are often left to assume the proceedings of the court without prior priming. There are significant possibilities of cognitive overload in witnesses leading to unfair and delayed proceedings.

The sheer impact and consequence of a witness’s contextually changing behaviour prompted a deeper investigation into the problem and work on cost effective and simple solutions to bring about an incremental positive change.

V. THE SYNTHESIS

The Proposed Solution - The triggers for both behaviour chains being vulnerability due to the spatial aspects of the witness box and the lack of prior priming respectively were tackled using two distinct interventions. The first proposed solution occurs during the testimony of the witness, to curb the witness’s unobstructed line-of-sight at the public which amplifies the spotlight effect. The second solution takes place in the waiting area of the Family Courtroom, to aid the witness in navigating an unfamiliar environment inside.

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These solutions were ideated to be low-cost, cognitive nudges that fit into the rules of the Court.

The spatial arrangements in open courtrooms in India, create the apt conditions for sustained social judgement and scrutiny, both real and perceived, that enhances cognitive overload due to audience gaze in witnesses. The proposed design intervention is adjusting the lighting such that the area above the public seating area is dimmed/switched off whereas the remaining portions of the courtroom are well lit. This subtle difference in lighting creates a “fourth wall” illusion leading the witness to imagine a visual barrier between themselves and the public. This solution not only fits within the standard procedure but also reduces the witness’s peripheral visibility. This intervention was tested out in simulations with design students and university staff and debriefing interviews post the simulation confirmed that participants experienced heightened anxiety and social judgement initially and increased confidence and reduced consciousness of the public post the lighting implementation. Both the cost and implications of this solution are minimal requiring no additional structural/infrastructure changes beyond electrical adjustments.

For a significant proportion of witnesses, the inside of a courtroom is an unfamiliar environment without prior exposure. The procedures in the courtroom, due to the absence of information, are constructed from their existing perceptions leading to compounding anticipatory anxiety. This increases their cognitive load even before stepping into the witness box. To counter this, the proposed intervention is a single-sided visual manual made available to the witnesses in the waiting area before their cross-examinations. The manual was designed in Gujarati which is also the primary language of most litigants at the Family Court at Gandhinagar. It consists of five steps that detail the code of conduct and reinforce that the court is a trustworthy environment through simple visuals and no legal jargon.

The participants in the simulation were given the manual while waiting their turn to participate. Debriefing interviews post the simulations confirmed the positive shift in how participants perceived the task about to be undertaken. Participants who read and understood the manual reported that they were less intimidated by the authority than they had anticipated. Enhanced confidence was reflected in the participants’ composure during the simulation. This manual is a low-cost and accessible solution which can be adapted in the vernacular languages of India across all states.

It is important to note that these solutions are not standalone. They target different phases in a witness’s journey and experience in the Family Court. The priming manual reduces anxiety in the waiting area before the testimony and the lighting adjustments reduce social scrutiny and distractions during the cross examination. Sequentially, these two interventions form a subtle yet strong framework that accompanies the witness from the waiting area to the witness box, targeting the specific points at which situational witness behaviours arise.

Worldview - The witness while on the witness box experiences a “spotlight” effect. The spotlight effect is a cognitive bias that creates the illusion of being under a spotlight increasing one’s consciousness of judgement, misremembrance and cognitive overload. It is a combination of isolation and judgement by an authority or panel that makes the situation quite distinct. While this effect is quite prominent in the legal space, it is also prevalent in academic contexts, hospitals, corporate presentations and media exposure. Each of these contexts has varying stakes, stakeholders, social formality, and authority. And in each context, the spotlight effect emerges as the fundamental basis of human cognition under pressure and observation.

Viva voce, quizzes and competitions are often designed to give unique and deeper insights into a student’s knowledge and thinking. These are also anxiety provoking and high pressure situations in the presence of authority and audience. The relationship between anxiety and academic/extracurricular activities is complex and is dependent on multiple varying factors. The

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spotlight effect and other cognitive biases could potentially lead to unsatisfactory outcomes resulting in reduced confidence and self esteem proving to be detrimental to one's journey.

Teaching hospitals conduct in-patient clinical grand rounds wherein each departmental unit consisting of a professor, associate professors, assistant professors, lecturers, senior residents, post graduates, and junior residents examine patients in detail. Medical reasons involve private and personal information and patients tend to seek reassurance regarding their health issues. During these rounds the patient who is already under medical duress is pushed into a socially vulnerable position where fear of social judgment is heightened. This could possibly lead to reluctance towards seeking further diagnosis and treatment.

Presentations in the corporate world are often presented on stages to senior leadership and in the presence of clients and an audience. The stage adds more beyond just being watched. Distractions from the audience can cause the presenter to lose their focus and break their thought flow. This snowballs into memory distortion and inconsistent speech. This increased performance anxiety leads to under-confidence and a tendency to seek clues to continue their train of thought.

Press conferences often have hostile questions thrown at the speaker. Under the eyes of the media, in high stake situations where one's words are recorded on-the-go, interruptions can prove costly. Speakers experience cognitive overload and bounded rationality biases that lead to inconsistent statements and anxiety. In matters of high-stakes such as those of national importance, inconsistent statements would have dire consequences when broadcast.

Across all five spaces and contexts of courtrooms, academic examinations, teaching hospitals, corporate presentations and media exposures, there exists a consistent structure. An individual is placed in a position where verbal exchanges with stakes take place in the presence of authority and an observing audience. In most cases, the individuals are also not primed to navigate the pressure of the situation. The combination of performance, authority, audience and lack of preparedness induces the spotlight effect.

In the Family Courtroom, the consequences of the individual's responses are at the highest possible stake as any witness who corrects their statements or is influenced by verbal or visual cues during the cross-examination risks altering the outcomes of cases that impact everyone involved in the case. As such, reduction of the spotlight effect and inducing a cognitive barrier in courtrooms is of the utmost importance. However, the proposed solutions could also be tailored to aid stakeholders in navigating other unfamiliar environments and possibly increase their confidence, focus and induce perceived and cognitive barriers.

VI. THE CHALLENGE

Challenges during the research can be broadly classified into two categories - rigidity in courtroom rules and its bureaucratic processes and data collection and analysis.

Court spaces do not allow for exceptions or variations. The same system that has been followed from the past couple of decades continues to be followed proving it difficult to address the challenges of adjournments, advocate behaviours, case preparedness, insufficient infrastructure and information presentation.

The process of slow ethnography from the lens of a visual thinker as opposed to that of an analytical thinker initially hindered the process of data collection. Identifying key observations in every action, differentiating an observation stemming from a behaviour versus a habit, recognizing the triggers for a behaviour, articulating and quantifying the observations right and the right observations, analyzing the articulated behaviours through different lenses and probing for emerging patterns were design research challenges. These challenges were addressed through multiple research briefings, debriefings,

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and simulation designs.

VII. THE RESIDUE

The proposed solutions open up wider questions on how spatial designs elicit adaptability in humans through behaviour alterations. These adaptabilities even when normalized by society do not necessarily reflect a positive change. Would increasing the lighting on witnesses on the witness box, cues placed on the witness box for micro-orientation or rhythmic breathing invoke a similar sense of safety and confidence in witnesses? Which behaviours would a witness exhibit on being seated at the witness box or on stepping on a raised witness box? How would efficient information design and prior courtroom experiential awareness affect a witness?

These questions and this study not only provoke thoughts on improving the overall process design, comfort, and protection in the Indian judicial system but also stress on the impact of solving smaller barriers and the systemic level changes that the solutions could snowball into. The understanding of human behaviour as the root cause of cascading problems that could be addressed through well-thought out nudges transforms deviances into sustainable and long-term changes.

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